Multithreaded algorithms: exercises

27-E01 Work, span, and processor lower bounds

For W=720, D=24, find lower and greedy upper time bounds at P=8,16. Do these metrics alone prove no benefit above30 processors? For 697 independent tasks followed by a23-node chain, prove exact optimum times at P=30,31.

27-E02 Recursive matrix work and dependencies

Eight order-n/2 products run simultaneously, then combine through nested parallel loops. Find work, span, and parallelism. Why cannot8 in 8W(n/2) be absorbed away? Where would eight contributions using += on four common C blocks conflict?

27-E03 Worst-case parallel-merge span

Derive child size$\leq$3N/4 from the longer-run median and use binary-search cost to prove O($\log^2$N) span. Give a matching lower-bound family. If the other run always stays empty, must span still be Θ($\log^2$N)?

27-E04 Scheduling nine Fibonacci strands

Build every dependency of P-FIB(3) in the three-strand model. Find W, D, parallelism, and a time-step greedy schedule for P=3. Prove its optimality. Derive exact general Wn and Dn for n$\geq$2.

27-E05 Lost updates and two repairs

Starting x=0, two tasks implement x=x+1 with load/add/store. Give a legal sequentially consistent interleaving ending at 1 and explain why final sync cannot repair it. Compare dependencies and span of atomic addition versus separate writes followed by reduction for n terms.

27-E06 Reduction for rectangular matrices

For positive p, q, r and A of size p$\times$q, B of size q$\times$r, give race-free multiplication with valid bounds even when any dimension is 1. Specify full dot-product recursion and outer ownership, then derive work, span, and space under explicit arithmetic/allocation models.

Multithreaded algorithms: exercises (continued)

27-E07 Longer-run swaps and tie rules

Merge A=[1a], B=[1b,2c], where a, b, c identify original order. Does swapping the longer run first and using key-only lower_bound preserve stability? Trace pivots and destinations. Repair using(key, original position) and give the general reason.

27-E08 Output offsets and aliasing

Apply P-MERGE to T=[1a,4b,6c,9d | 2e,3f,6g,8h] with o=3. Find root m, j, q, child output intervals, and final O[3..11). If output instead aliases T with o=0, which unread input does the first root write destroy?

27-E09 General balanced-tree charging

A recursion on N records removes one unique record per nonempty call. Its disjoint children each have size$\leq \rho m$ for fixed$0<\rho<1$; nonrecursive cost is O(log(m+1)). What child-count assumption is needed to include empty-call costs? With it, prove O(N) work by distributing charges. Give a chain counterexample when ρ may approach1 with m.

27-E10 One input family must realize worst sort span

Why does summing independently worst merge spans not establish a sort lower bound? Construct the n=16 bit-reversed key order, give the root's sorted runs, and explain recursive similarity to obtain worst span $\Theta(\log^3 n)$.

27-E11 Reversing the work/span preference

Correct algorithms A, B have(W, D)=(2048,8),(1536,32). Using only work/span lower bounds and greedy W/P+D upper bounds, find integer-P ranges guaranteeing B faster and A faster. How should the intervening range be described?

27-E12 Serial spawning differs from parallel for

Batch n unit iterations into groups of g, with ceil(n/g) groups. Compare work/span of an ordinary loop spawning groups versus balanced PAR-FOR. Which g gives the first method's best asymptotic span? Count the parent's task-issuing chain.

Multithreaded algorithms: exercises (continued)

27-E13 Matrix multiplication with less temporary storage

After parallel zeroing of C, design ACC(C, A, B) for C←C+AB without an n×n temporary. Use two synchronized phases per C block, with four different blocks concurrent within a phase. Give full recursion, correctness, work/span, space, and comparison to full eight-way expansion.

27-E14 Two meanings of average random-DAG speedup

Equally likely DAGs have respectively W=100, D=10, T10=10 and W=100, D=100, T10=100. Compare E[W/T10] and E[W]/E[T10], and give realizable DAGs. Derive an expected-time bound from the pointwise greedy theorem. Must W and D be independent?

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

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https://github.com/xinhi6716/computer-science-self-study

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